

Chapter 5 exercises — Analytic Combinatorics

26 exercises, extracted from the textbook PDF.

Caveats. (1) Statements are auto-extracted; `fi` / `fl` ligatures dropped by the PDF text layer were repaired by a cleanup pass (a rare odd gap may remain). (2) A $\triangle$ marks exercises whose statement includes a **displayed formula** that did not survive extraction — read those in the book at the cited page. The book PDF is always authoritative.

#	p.	Exercise
5.1	227	How many bitstrings of length N have no 000?
5.2	227	How many bitstrings of length N have no 01?
5.3	228	Let U be the set of binary trees with the size of a tree defined to be the total number of nodes (internal plus external), so that the generating function for its counting sequence is $U(z) = z + z^3 + 2z^5 + 5z^7 + 14z^9 + \dots$. Derive an explicit expression for $U(z)$.
5.4	228	Define a "superleaf" in a binary tree to be an internal node whose four grandchildren are all external nodes. What fraction of binary trees with N nodes have no superleaves? From Figure 5.2, the answer for $N = 1, 2, 3$, and 4 is 0, 0, 4/5, and 6/7, respectively.
5.5	229	Let B be defined as the collection of all finite subsets of A . If $A(z)$ and $B(z)$ are the OGFs of A and B , show that $B(z) =$
5.6	229	Let B be defined as the collection of all finite multisets of A (subsets with repetitions allowed). If $A(z)$ and $B(z)$ are the OGFs of A and B , show that $B(z) =$
5.7	240	Derive an EGF for the number of permutations whose cycles are all of odd length.
5.8	240	Derive an EGF for sequences of cycles.
5.9	240	Derive an EGF for cycles of sequences.
5.10	242	Extend the proof of
5.11	242	Extend the proof of
5.12	246	Confirm that the average number of 1 bits in a random bitstring is $N/2$ by computing $B_u(z, 1)$.
5.13	246	What is the average number of 1 bits in a random bitstring of length N having no 00?
5.14	246	What is the average number of cycles in a random derangement?
5.15	246	Find the average number of internal nodes in a binary tree of size n with both children internal.
5.16	246	Find the average number of internal nodes in a binary tree of size n with one child internal and one child external.
5.17	246	Find an explicit formula for $T(z, u)$ and compute the variance of the number of leaves in binary trees.

#	p.	Exercise
5.18	247	Use Taylor's theorem to find $[z^N]$
5.19	248	Use eorem $\sqrt{4.1}$ to show that the number of bitstrings having no occurrence of 00 is $\sim \phi^N / 5$.
5.20	248	Find an approximation for the number of bitstrings having no occurrence of 01.
5.21	248	Prove that there exists a constant C such that $\sqrt{[z^n] \exp(z/(1-z))} = O(\exp(Cn))$.
5.22	248	Establish similar bounds for the OGF of integer partitions $\prod_{k \geq 1} [z^n] (1 - z^k)^{-1}$.
5.23	252	Show that the probability $\sqrt{\text{that all the cycles are of odd length in a random permutation of length } N}$ is $1/\pi^{N/2}$ (see Exercise 5.7).
5.24	252	Give a more precise version of series to three terms.
5.25	253	Show that $[z^n] f(z) \ln$
5.26	253	Give a transfer theorem like $[z^n] f(z)$